Vegetated buffer and flower strips

Spatial scale: landscape

About this sustainable management strategy

Vegetated buffer strips and flower strips involve the establishment and management of permanent or semi-permanent vegetation within or adjacent to vineyards to reduce environmental pressures and enhance ecosystem functioning. They are typically located along vineyard margins, terrace embankments, waterways, roadsides, and other non-cultivated areas where they act as ecological buffers between vineyards and surrounding landscapes.
These strips are composed of diverse plant communities, including native grasses, herbs, and flowering species, which can be established through sowing regional seed mixtures or by promoting the natural development of spontaneous vegetation through extensive management. Depending on their location and design, buffer strips can intercept sediment, nutrients, and pesticide residues transported by surface runoff, thereby reducing the risk of water contamination and soil loss. When implemented as hedgerows or other woody landscape elements, they can also reduce wind erosion and pesticide drift, particularly when positioned perpendicular to prevailing wind directions.
In addition to their protective functions, vegetated buffer and flower strips provide habitat, food resources, and ecological corridors for pollinators, natural enemies of pests, and other wildlife. By increasing habitat connectivity and structural diversity within agricultural landscapes, they contribute to biodiversity conservation and support ecosystem resilience under changing environmental conditions.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling, and organic matter content*)
- Erosion control *(slope stabilisation and prevention of run-off)*
- Carbon sequestration
- Water retention *(reduced surface runoff)*
- Water regulation *(improved infiltration, retention, and reduced nutrient leaching)*
- Biodiversity enhancement (*above- and below-ground*)
- Natural pest control *(support for beneficial insects and natural enemies)*
- Climate regulation (*local microclimate regulation, reduced temperature extremes*)
- Pollination services *(food and habitat resources for pollinators)*
- Landscape aesthetics

Considerations for successful implementation

Appropriate establishment and maintenance
Successful buffer strips require careful establishment and ongoing management. Selecting suitable native or locally adapted species, ensuring good establishment, and carrying out periodic maintenance can help maximise biodiversity benefits while preventing the dominance of undesirable species.

Site-specific design
The effectiveness of vegetated buffer strips depends on their width, vegetation composition, and placement within the landscape. Adapting their design to local conditions such as slope, soil

type, climate, and runoff pathways can enhance their capacity to reduce erosion, retain pollutants, and support biodiversity.

Regular monitoring and adaptive management

Regular monitoring helps ensure that buffer strips continue to deliver their intended benefits. Adaptive management, including adjustments to vegetation composition and maintenance practices, can help maintain ecological functioning, support beneficial organisms, and minimise potential pest or weed-related issues.